

Workshop Practice Lab:

Workshop Practice Lab is a practical course that introduces students to the basic techniques and tools used in various engineering workshops. The course is designed to provide hands-on experience in the use of workshop tools and equipment, and to develop skills in the fabrication of simple engineering components. Some of the key topics covered in the Workshop Practice Lab include:

1. **Safety:** Students learn about the importance of safety in the workshop, including the use of personal protective equipment and safe working practices.
2. **Hand Tools:** Students are introduced to the use of common hand tools such as hammers, screwdrivers, pliers, and wrenches.
3. **Measuring Instruments:** Students learn how to use measuring instruments such as vernier calipers, micrometers, and dial gauges to take accurate measurements.
4. **Machine Tools:** Students gain experience in the use of machine tools such as lathes, milling machines, and drilling machines.
5. **Welding:** Students are introduced to the basics of welding, including the use of welding equipment and the different types of welding processes.
6. **Sheet Metal Work:** Students learn how to cut, bend, and shape sheet metal to create simple components.
7. **Fitting:** Students learn how to fit and assemble mechanical components using techniques such as filing, sawing, and drilling.

The Workshop Practice Lab provides a solid foundation for students who wish to pursue further studies or careers in engineering and related fields. It is an essential course for anyone who wants to gain practical skills and experience in the use of workshop tools and equipment.

S. No. Mechanical Workshop Duration

1. **Introduction to Mechanical Workshop:** Mechanical workshop is a place where various mechanical operations are performed using different tools and machines. The duration of a mechanical workshop can vary depending on the scope and objectives of the workshop.
2. **Factors affecting the duration of a mechanical workshop:** The duration of a mechanical workshop can be affected by several factors such as the number of participants, the complexity of the tasks to be performed, the availability of tools and machines, and the level of expertise of the participants.
3. **Typical duration of a mechanical workshop:** A typical mechanical workshop can last anywhere from a few hours to several days. Shorter work-

shops may focus on a specific topic or skill, while longer workshops may cover a broader range of topics and provide more hands-on experience.

4. Importance of planning the duration of a mechanical workshop: It is important to carefully plan the duration of a mechanical workshop to ensure that all objectives are met and that participants have enough time to learn and practice new skills. A well-planned workshop can help participants to achieve their goals and improve their skills in a more efficient and effective manner.
5. Conclusion: The duration of a mechanical workshop can vary depending on several factors. Careful planning and consideration of these factors can help to ensure that the workshop is successful and meets the needs of all participants.

1 Introduction to Mechanical workshop material, tools and machines

3 Hrs

A mechanical workshop is a place where various materials, tools, and machines are used to manufacture, repair, or maintain mechanical systems. In this section, we will introduce some common materials, tools, and machines found in a mechanical workshop.

Materials

- **Metals:** Metals such as steel, aluminum, and copper are commonly used in mechanical workshops due to their strength, durability, and conductivity.
- **Plastics:** Plastics are also commonly used in mechanical workshops due to their versatility and low cost. Some common types of plastics used in mechanical workshops include polyethylene, polypropylene, and PVC.
- **Wood:** Wood is often used in mechanical workshops for its strength, durability, and aesthetic appeal.

Tools

- **Hand tools:** Hand tools such as hammers, screwdrivers, and pliers are commonly used in mechanical workshops for tasks such as fastening, cutting, and shaping.
- **Power tools:** Power tools such as drills, saws, and grinders are also commonly used in mechanical workshops for tasks that require more power and speed.
- **Measuring tools:** Measuring tools such as rulers, calipers, and micrometers are used in mechanical workshops to ensure accuracy and precision in the manufacturing process.

Machines

- **Lathes:** Lathes are machines used in mechanical workshops to shape materials such as metal, wood, or plastic by rotating them against a cutting tool.
- **Milling machines:** Milling machines are used in mechanical workshops to cut and shape materials using a rotating cutting tool.
- **Drilling machines:** Drilling machines are used in mechanical workshops to create holes in materials using a rotating drill bit.

In summary, a mechanical workshop is a place where various materials, tools, and machines are used to manufacture, repair, or maintain mechanical systems. Understanding the common materials, tools, and machines found in a mechanical workshop is essential for anyone working in or studying the field of mechanics.

To study layout, safety measures and different engineering materials (mild steel, medium carbon steel, high carbon steel, high speed steel and cast iron etc) used in workshop.

- **Layout:** The layout of a workshop refers to the arrangement of equipment, machinery, and workstations in a manner that promotes efficient workflow and safety. A well-designed layout can improve productivity, reduce accidents, and minimize waste.
- **Safety Measures:** Safety measures in a workshop include the use of personal protective equipment (PPE) such as safety glasses, gloves, and earplugs, as well as the implementation of safety procedures and protocols. These measures are designed to prevent accidents and injuries, and to protect workers from hazards such as flying debris, loud noise, and dangerous chemicals.
- **Engineering Materials:** Different engineering materials are used in a workshop, depending on the specific needs of the project. Some common materials include:
 - **Mild Steel:** Mild steel is a type of carbon steel with a low carbon content. It is strong, ductile, and easy to weld, making it a popular choice for construction and manufacturing.
 - **Medium Carbon Steel:** Medium carbon steel has a higher carbon content than mild steel, which makes it stronger and harder. It is often used for applications that require high strength and durability, such as gears and axles.
 - **High Carbon Steel:** High carbon steel has an even higher carbon content than medium carbon steel, which makes it very hard and strong. It is often used for applications that require high wear resistance, such as cutting tools and knives.
 - **High Speed Steel:** High speed steel is a type of tool steel that

is designed to withstand high temperatures and high cutting speeds. It is often used for making cutting tools such as drill bits and saw blades.

- **Cast Iron:** Cast iron is a group of iron-carbon alloys with a carbon content greater than 2%. It is strong, brittle, and resistant to wear, making it a popular choice for applications such as engine blocks and pipes.

These are just a few examples of the many different engineering materials that can be used in a workshop. The specific materials chosen will depend on the requirements of the project and the preferences of the engineer or technician.

To study and use of different types of tools, equipment, devices & machines used in fitting, sheet metal and welding section

Fitting, sheet metal, and welding are important processes in the manufacturing industry. These processes require the use of various tools, equipment, devices, and machines to achieve the desired results. Below is a list of some of the common tools, equipment, devices, and machines used in fitting, sheet metal, and welding section:

1. **Fitting tools:** These include tools such as hammers, chisels, files, hack-saws, and hand reamers. These tools are used for cutting, shaping, and finishing metal parts.
2. **Sheet metal tools:** These include tools such as snips, shears, punches, and dies. These tools are used for cutting, bending, and shaping sheet metal.
3. **Welding equipment:** This includes equipment such as welding machines, welding torches, and welding electrodes. These are used for joining metal parts together through the process of welding.
4. **Measuring and marking tools:** These include tools such as rulers, calipers, and scribes. These tools are used for measuring and marking metal parts to ensure accuracy and precision.
5. **Machines:** These include machines such as lathes, milling machines, and drill presses. These machines are used for cutting, shaping, and drilling metal parts.

It is important to note that the specific tools, equipment, devices, and machines used in fitting, sheet metal, and welding section may vary depending on the specific task and the materials being used. It is important to use the appropriate tools and equipment for the task at hand to ensure safety and efficiency.

To determine the least count of Vernier calliper, vernier height gauge, micrometer (Screw gauge) and take different reading over given metallic pieces using these instruments.

- **Vernier caliper:** The least count of a Vernier caliper is determined by the difference between the value of one main scale division and the value of one Vernier scale division. The formula for calculating the least count of a Vernier caliper is given by: $\text{Least Count (LC)} = (\text{Value of 1 Main Scale Division}) - (\text{Value of 1 Vernier Scale Division})$.
- **Vernier height gauge:** The least count of a Vernier height gauge is determined in the same way as that of a Vernier caliper. The formula for calculating the least count of a Vernier height gauge is given by: $\text{Least Count (LC)} = (\text{Value of 1 Main Scale Division}) - (\text{Value of 1 Vernier Scale Division})$.
- **Micrometer (Screw gauge):** The least count of a micrometer is determined by the pitch of the screw, which is the distance moved by the spindle per revolution. The formula for calculating the least count of a micrometer is given by: $\text{Least Count (LC)} = (\text{Pitch of the Screw}) / (\text{Number of Divisions on the Thimble})$.

To take different readings over given metallic pieces using these instruments, the following steps can be followed:

1. Place the metallic piece between the jaws of the instrument.
2. Adjust the position of the jaws until they are in contact with the metallic piece.
3. Read the main scale and the Vernier scale (for Vernier caliper and Vernier height gauge) or the thimble scale (for micrometer) to obtain the measurement.
4. Repeat the process for different metallic pieces and record the readings.

Machine Shop (3 hours)

A machine shop is a facility where machining is performed. Machining is the process of removing material from a workpiece to create a desired shape or surface. Here are some key points to know about machine shops:

1. **Machining equipment:** Machine shops typically contain a variety of equipment used for cutting, shaping, drilling, and finishing metal and other materials. Some common types of equipment found in machine shops include lathes, milling machines, drill presses, and grinders.
2. **Skilled workers:** Machine shops employ skilled workers, such as machinists and tool and die makers, who are trained to operate the equipment and produce precision parts.
3. **Types of work:** Machine shops can produce a wide range of parts, from small components used in electronics to large, complex parts used in aerospace and other industries. They may also perform repair and maintenance work on machinery and equipment.

4. **Safety:** Safety is a top priority in machine shops, as the equipment can be dangerous if not used properly. Workers must follow strict safety protocols and wear appropriate personal protective equipment.
5. **Quality control:** Machine shops must adhere to strict quality control standards to ensure that the parts they produce meet the required specifications. This may involve using precision measuring instruments to verify the dimensions and tolerances of the parts.
6. **Computer-aided manufacturing:** Many machine shops now use computer-aided manufacturing (CAM) software to control the equipment and automate the production process. This can increase efficiency and accuracy, and reduce the need for manual labor.

Demonstration of working, construction and accessories for Lathe machine Perform operations on Lathe - Facing, Plane Turning , step turning, taper turning, threading, knurling and parting.

A lathe is a machine tool that rotates a workpiece about an axis of rotation to perform various operations such as cutting, sanding, knurling, drilling, deformation, facing, and turning. The lathe is one of the oldest and most important machine tools.

The main parts of a lathe machine are:

1. **Headstock:** The headstock is located on the left side of the bed and serves as a housing for the driving pulleys and back gears. It also contains the main spindle and the change gears.
2. **Tailstock:** The tailstock is located on the right side of the bed and is used to support the other end of the workpiece.
3. **Bed:** The bed is the main supporting structure of the lathe. It is a long, flat base that is mounted on legs or a cabinet.
4. **Carriage:** The carriage is a movable platform that supports the cutting tool and moves along the bed.
5. **Cross slide:** The cross slide is mounted on the carriage and moves perpendicular to the bed.
6. **Compound rest:** The compound rest is mounted on the cross slide and can be rotated to any angle.
7. **Tool post:** The tool post is mounted on the compound rest and holds the cutting tool.
8. **Lead screw:** The lead screw is a long threaded rod that runs the length of the bed and is used to move the carriage.
9. **Feed rod:** The feed rod is a long rod that runs parallel to the lead screw and is used to move the carriage.

The lathe can perform various operations such as:

- **Facing:** Facing is the process of machining the end of a workpiece to produce a flat surface.
- **Plane Turning:** Plane turning is the process of machining a cylindrical surface.
- **Step Turning:** Step turning is the process of machining a stepped cylindrical surface.
- **Taper Turning:** Taper turning is the process of machining a conical surface.
- **Threading:** Threading is the process of cutting a screw thread on a workpiece.
- **Knurling:** Knurling is the process of producing a rough surface on a workpiece.
- **Parting:** Parting is the process of cutting off a piece of the workpiece.

These are the basic operations that can be performed on a lathe machine. There are many other operations that can be performed on a lathe, depending on the accessories and attachments used. Some of these operations include drilling, boring, reaming, and tapping.

Fitting Shop (3 Hours)

1. A fitting shop is a workshop where the process of fitting, assembling, and finishing of machine parts takes place.
2. The fitting shop is equipped with various hand tools, power tools, and measuring instruments to carry out the fitting process.
3. The fitting process involves shaping, cutting, drilling, filing, and grinding of metal parts to achieve the desired shape and size.
4. The fitting shop is an essential part of any manufacturing industry, as it is responsible for the final assembly and finishing of machine parts.
5. The fitting shop requires skilled workers who are trained in the use of various tools and instruments to carry out the fitting process accurately and efficiently.
6. The duration of the fitting process can vary depending on the complexity of the machine parts being assembled and the level of precision required.
7. In this context, 3 hours may be allocated for the fitting process in the fitting shop. This time may be used for the fitting, assembling, and finishing of machine parts.

1. Practice marking operations

- Marking operations are used to identify and label objects, locations, or features.

- These operations can be performed using various tools such as markers, pens, chalk, or paint.
- Marking operations are commonly used in industries such as construction, manufacturing, and logistics.
- To practice marking operations, one can start by selecting the appropriate tool for the task at hand.
- Next, the user should practice making clear and legible marks on various surfaces such as paper, wood, metal, or concrete.
- It is important to follow any safety guidelines or regulations when performing marking operations.
- With practice, the user can improve their accuracy and speed when performing marking operations.

2. Preparation of U or V -Shape Male Female Work piece which contains: Filing, Sawing, Drilling, Grinding.

The preparation of U or V-shaped male and female work pieces involves several steps, including filing, sawing, drilling, and grinding. These steps are necessary to achieve the desired shape and fit of the work pieces.

1. **Filing:** Filing is the process of removing material from the work piece using a file. This is done to smooth out rough edges and to shape the work piece to the desired form.
2. **Sawing:** Sawing is the process of cutting the work piece to the desired size and shape using a saw. This is typically done using a band saw or a hacksaw.
3. **Drilling:** Drilling is the process of creating holes in the work piece using a drill. This is necessary for the assembly of the male and female work pieces.
4. **Grinding:** Grinding is the process of removing material from the work piece using an abrasive wheel. This is done to smooth out rough surfaces and to achieve a precise fit between the male and female work pieces.

These steps must be performed carefully and accurately to ensure that the male and female work pieces fit together properly. Proper preparation of the work pieces is essential for the success of the final product.

4 Carpentry Shop 3 Hrs

Carpentry is a skilled trade and a craft in which the primary work performed is the cutting, shaping and installation of building materials during the construction of buildings, ships, timber bridges, concrete formwork, etc. Carpenters traditionally worked with natural wood and did the rougher work such as framing, but today many other materials are also used and sometimes the finer trades of cabinetmaking and furniture building are considered carpentry.

A carpentry shop is a workshop where carpenters work. It is typically equipped with tools and machinery for cutting, shaping, and joining wood. Some common tools and machinery found in a carpentry shop include:

1. Hand tools such as saws, hammers, chisels, and planes.
2. Power tools such as circular saws, jigsaws, and drills.
3. Stationary machinery such as table saws, band saws, and jointers.
4. Measuring and marking tools such as rulers, squares, and levels.

In a 3-hour session at a carpentry shop, one could expect to learn and practice basic carpentry skills such as measuring and marking, cutting and shaping wood, and joining pieces together. One could also work on a small project such as building a birdhouse or a picture frame.

It is important to note that safety is a top priority in a carpentry shop. Proper safety equipment such as eye and ear protection should be worn at all times, and all tools and machinery should be used in accordance with the manufacturer's instructions and safety guidelines. Additionally, it is important to keep the shop clean and organized to prevent accidents.

Study of Carpentry Tools, Equipment and different joints. Making of Cross Half lap joint, Half lap Dovetail joint and Mortise Tenon Joint

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Carpentry tools and equipment include: - Measuring tools: tape measure, combination square, framing square, level - Cutting tools: hand saw, circular saw, jigsaw, miter saw - Shaping tools: plane, chisel, router - Fastening tools: hammer, nail gun, screwdriver, drill - Clamping tools: bar clamp, C-clamp, pipe clamp

Different types of joints used in carpentry include: - Butt joint: the simplest joint where two pieces of wood are butted together - Lap joint: a joint where one piece of wood overlaps the other - Dovetail joint: a joint where the ends of two pieces of wood are cut in a series of interlocking fingers - Mortise and tenon joint: a joint where a tenon on one piece of wood fits into a mortise on the other

Cross Half Lap Joint: - A cross half lap joint is a type of lap joint where two pieces of wood cross each other and half of the material is removed from each piece so that they fit together. - To make a cross half lap joint, first mark the center of each piece of wood where they will cross. - Use a saw to cut halfway through the thickness of each piece of wood at the center mark. - Use a chisel

to remove the waste material and create a flat surface. - Fit the two pieces together so that they overlap at the center and are flush on both sides.

Half Lap Dovetail Joint: - A half lap dovetail joint is a type of lap joint where one piece of wood overlaps the other and the ends are cut in a series of interlocking fingers. - To make a half lap dovetail joint, first mark the center of each piece of wood where they will overlap. - Use a saw to cut halfway through the thickness of each piece of wood at the center mark. - Use a dovetail saw to cut the fingers on the ends of each piece of wood. - Use a chisel to remove the waste material and create a flat surface. - Fit the two pieces together so that the fingers interlock and the joint is flush on both sides.

Mortise and Tenon Joint: - A mortise and tenon joint is a type of joint where a tenon on one piece of wood fits into a mortise on the other. - To make a mortise and tenon joint, first mark the location of the mortise on one piece of wood and the location of the tenon on the other. - Use a drill to remove most of the waste material from the mortise. - Use a chisel to square up the sides of the mortise and create a flat bottom. - Use a saw to cut the tenon to the correct size and shape. - Fit the tenon into the mortise and make any necessary adjustments to ensure a tight fit.

5 Welding Shop 6 Hrs

Welding is a fabrication process that joins two or more parts by heating the surfaces to their melting points and applying pressure to fuse them together. Welding is used in a variety of industries, including construction, manufacturing, and repair.

A welding shop is a facility where welding operations are performed. A typical welding shop may include the following equipment:

1. Welding machines: These are used to generate the heat and electrical current required for welding.
2. Welding torches: These are used to direct the heat and electrical current to the area being welded.
3. Welding electrodes: These are used to conduct the electrical current to the area being welded.
4. Protective equipment: This includes items such as welding helmets, gloves, and aprons to protect the welder from heat, sparks, and ultraviolet radiation.
5. Hand tools: These include items such as hammers, pliers, and wire brushes for preparing and finishing the weld.

A 6-hour welding shop session may involve a variety of activities, including:

1. Preparing the materials to be welded: This may involve cutting, shaping, and cleaning the materials.
2. Setting up the welding equipment: This may involve selecting the appropriate welding machine, torch, and electrode for the job.

3. **Welding the materials:** This involves using the welding equipment to join the materials together.
4. **Finishing the weld:** This may involve grinding, sanding, or polishing the weld to improve its appearance and strength.
5. **Cleaning up:** This involves disposing of any waste materials and returning the equipment to its proper storage location.

Welding is a skilled trade that requires training and practice to master. A 6-hour welding shop session can provide valuable hands-on experience for those learning the trade.

Introduction to BI standards and reading of welding drawings

BI standards refer to the standards set by the British Institute of Standards for welding and fabrication. These standards provide guidelines for the design, execution, and inspection of welded joints and structures.

Reading welding drawings is an essential skill for anyone involved in the welding and fabrication industry. Welding drawings provide detailed information about the dimensions, materials, and welding processes to be used in the fabrication of a welded structure.

In order to read welding drawings, one must be familiar with welding symbols, dimensions, and welding terminology. Welding symbols are used to indicate the type of weld, its size, and its location on the drawing. Dimensions provide information about the size and shape of the welded structure, while welding terminology refers to the specific welding processes and techniques used.

Practice of Making following operations

1. **Butt Joint:** A butt joint is a type of joint where two pieces of material are joined end to end. This type of joint is commonly used in welding and can be created using various welding techniques such as TIG or MIG welding.
2. **Lap Joint:** A lap joint is a type of joint where two pieces of material are overlapped and joined together. This type of joint is commonly used in welding and can be created using various welding techniques such as TIG or MIG welding.
3. **TIG Welding:** TIG welding, also known as Gas Tungsten Arc Welding (GTAW), is a welding process that uses a non-consumable tungsten electrode to produce the weld. This type of welding is commonly used for welding thin materials and for creating precise welds.
4. **MIG Welding:** MIG welding, also known as Gas Metal Arc Welding (GMAW), is a welding process that uses a consumable wire electrode to produce the weld. This type of welding is commonly used for welding thicker materials and for creating strong welds.

These are some of the basic operations and techniques used in welding and fabrication. It is important to practice these techniques in order to become proficient in the art of welding.

6 Moulding and Casting Shop 6 Hrs

Moulding and casting are manufacturing processes used to produce complex shapes. These processes are commonly used in metalworking, but can also be used with other materials such as plastic and glass.

1. **Moulding** is the process of shaping a material by pouring or pressing it into a mould. The mould is typically made of a hard material, such as metal or ceramic, and is designed to have the shape of the desired final product. Once the material has been poured or pressed into the mould, it is allowed to cool and solidify, taking on the shape of the mould.
2. **Casting** is a similar process, but instead of pressing the material into the mould, it is melted and poured into the mould. The material is then allowed to cool and solidify, taking on the shape of the mould. Casting is commonly used with metals, but can also be used with other materials such as plastic and glass.
3. Both moulding and casting can be used to produce complex shapes that would be difficult or impossible to produce using other manufacturing processes. They are commonly used to produce parts for machinery, tools, and other products.
4. In a moulding and casting shop, workers will typically use specialized equipment to create the moulds and cast the materials. This can include furnaces for melting metal, presses for shaping materials, and other tools for preparing and finishing the final product.
5. The process of moulding and casting can take several hours, depending on the complexity of the shape and the material being used. In a 6-hour moulding and casting shop, workers will typically produce several parts or products during their shift.
6. Moulding and casting are important processes in many industries, including automotive, aerospace, and consumer goods. They allow manufacturers to produce complex shapes quickly and efficiently, making them an essential part of modern manufacturing.

Introduction to Patterns, Pattern Allowances, Ingredients of Moulding Sand and Melting Furnaces. Foundry Tools and Their Purposes Demo of Mould Preparation and Aluminum Casting Practice – Study and Preparation of Mould for Plastic

1. **Patterns:** A pattern is a model or replica of the object to be cast. It is used to create a cavity in the moulding sand, into which molten metal is

poured to produce a casting.

2. **Pattern Allowances:** Pattern allowances are small adjustments made to the pattern to compensate for changes that occur during the casting process. These include shrinkage allowance, draft allowance, machining allowance, and distortion allowance.
3. **Ingredients of Moulding Sand:** Moulding sand is a mixture of sand, clay, and water. The sand provides the structure for the mould, while the clay binds the sand particles together. Water is added to make the sand mixture pliable.
4. **Melting Furnaces:** Melting furnaces are used to melt the metal that will be poured into the mould. There are several types of melting furnaces, including cupola furnaces, induction furnaces, and electric arc furnaces.
5. **Foundry Tools:** Foundry tools are used to prepare the mould and handle the molten metal. These include moulding boxes, rammers, sprue cutters, and tongs.
6. **Demo of Mould Preparation and Aluminum Casting Practice:** In this demo, a mould is prepared using a pattern and moulding sand. Molten aluminum is then poured into the mould to produce a casting. The casting is then removed from the mould and any excess material is trimmed away.
7. **Study and Preparation of Mould for Plastic:** In this practice, a mould is prepared for casting plastic. The process is similar to that used for metal casting, but the materials and techniques used may be different.

7 CNC Shop 6 Hrs

1. CNC (Computer Numerical Control) is a manufacturing process in which pre-programmed computer software dictates the movement of factory tools and machinery.
2. A CNC shop is a facility that specializes in CNC machining, which involves the use of computer-controlled machines to produce precision parts and components.
3. A 6-hour shift in a CNC shop would involve operating and monitoring CNC machines, performing quality checks on produced parts, and performing routine maintenance on the machines.
4. CNC machining is used in a wide range of industries, including aerospace, automotive, and medical device manufacturing.
5. CNC machines can perform a variety of operations, including milling, turning, and drilling, and can work with a wide range of materials, including metals, plastics, and composites.
6. CNC machining offers high precision and repeatability, making it a popular choice for producing complex parts with tight tolerances.
7. To work in a CNC shop, one typically needs training in CNC machine operation and programming, as well as knowledge of machining processes and materials.

Study of main features and working parts of CNC machine and accessories that can be used. Perform different operations on metal components using any CNC machines

CNC (Computer Numerical Control) machines are automated machines that are capable of performing a wide range of operations on metal components. These machines are controlled by computer programs and can be used to perform tasks such as drilling, milling, turning, and grinding.

Some of the main features of CNC machines include:

1. High precision and accuracy: CNC machines are capable of producing components with a high degree of precision and accuracy, thanks to their computer-controlled operation.
2. Flexibility: CNC machines can be programmed to perform a wide range of operations, making them highly flexible and versatile.
3. High productivity: CNC machines can operate continuously for long periods of time, increasing productivity and efficiency.
4. Reduced human intervention: CNC machines require minimal human intervention, reducing the risk of human error and increasing safety.

Some of the main working parts of a CNC machine include:

1. The control panel: This is the interface between the operator and the machine, and is used to input commands and control the operation of the machine.
2. The spindle: This is the part of the machine that rotates and holds the cutting tool.
3. The cutting tool: This is the part of the machine that performs the actual cutting or shaping of the metal component.
4. The worktable: This is the part of the machine that holds the metal component in place during the machining process.

There are a wide range of accessories that can be used with CNC machines to enhance their capabilities and perform different operations on metal components. Some common accessories include:

1. Tool holders: These are used to hold the cutting tools in place during the machining process.
2. Workholding devices: These are used to hold the metal component in place during the machining process.
3. Coolant systems: These are used to cool the cutting tool and the metal component during the machining process, reducing the risk of damage and increasing the lifespan of the cutting tool.

4. **Probes:** These are used to measure the dimensions of the metal component and ensure that it is machined to the correct specifications.

CNC machines can be used to perform a wide range of operations on metal components, including drilling, milling, turning, and grinding. These operations can be performed with a high degree of precision and accuracy, thanks to the computer-controlled operation of the machine. By using the appropriate accessories and programming the machine correctly, it is possible to produce high-quality metal components using a CNC machine.

8 To prepare a product using 3D printing 3 Hrs

1. **Design the 3D model:** The first step in preparing a product using 3D printing is to design a 3D model of the product using computer-aided design (CAD) software. This model will serve as the blueprint for the 3D printer to create the physical product.
2. **Choose the right 3D printing technology:** There are several different 3D printing technologies available, each with its own advantages and disadvantages. The most common technologies are Fused Deposition Modeling (FDM), Stereolithography (SLA), and Selective Laser Sintering (SLS). Choose the technology that best suits the requirements of your product.
3. **Select the appropriate material:** The material used for 3D printing will depend on the chosen 3D printing technology and the desired properties of the final product. Common materials used in 3D printing include plastics, resins, and metals.
4. **Prepare the 3D printer:** Before starting the printing process, the 3D printer must be prepared according to the manufacturer's instructions. This may include loading the material, calibrating the printer, and setting the printing parameters.
5. **Start the printing process:** Once the 3D printer is prepared, the printing process can begin. The 3D printer will use the 3D model as a guide to build the product layer by layer.
6. **Post-processing:** After the printing process is complete, the product may require post-processing to achieve the desired finish. This may include removing support structures, sanding, or painting.
7. **Quality control:** Finally, the finished product should be inspected for quality and tested to ensure that it meets the desired specifications.

In summary, preparing a product using 3D printing involves designing a 3D model, choosing the right 3D printing technology and material, preparing the 3D printer, starting the printing process, post-processing, and quality control. The entire process can take around 3 hours, depending on the complexity of the product and the chosen 3D printing technology.

Total 33 Hrs

- Total 33 hours refers to the total amount of time allocated for a specific task or activity.
- It can be used in various contexts such as work, study, or leisure.
- It is important to manage time effectively to ensure that all tasks are completed within the given time frame.
- Time management techniques such as setting goals, prioritizing tasks, and avoiding distractions can help in utilizing the 33 hours effectively.
- It is also important to take breaks and rest to avoid burnout and maintain productivity.

Course Outcome:

- A course outcome is a statement that describes the knowledge, skills, and abilities that students are expected to have acquired by the end of a course.
- Course outcomes are used to assess the effectiveness of a course in achieving its intended goals.
- Course outcomes are typically written in measurable terms, using action verbs that describe what students should be able to do by the end of the course.
- Course outcomes should be aligned with the overall goals and objectives of the program or curriculum in which the course is offered.
- Course outcomes can be used to guide the development of course content, assessments, and instructional strategies.
- Course outcomes can also be used to communicate expectations to students and to provide a basis for evaluating student learning.
- Course outcomes should be reviewed and revised periodically to ensure that they remain relevant and up-to-date.

The students will be able to:

- CO1: Use various engineering materials, tools, machines and measuring equipment at a K3 Bloom's level.
- CO2: Perform machine operations in lathe and CNC machine at a K3 Bloom's level.
- CO3: Perform manufacturing operations on components in fitting and carpentry shop at a K3 Bloom's level.
- CO4: Perform operations in welding, moulding, casting and gas cutting at a K3 Bloom's level.
- CO5: Fabricate a job by 3D printing manufacturing technique at a K3 Bloom's level.

Reference Books:

Reference books are a type of non-fiction book that provides information on a specific subject. They are often used for research and study purposes. Some

common types of reference books include:

1. **Encyclopedias:** These books provide comprehensive information on a wide range of subjects. They are often organized alphabetically and contain entries on various topics.
2. **Dictionaries:** These books provide definitions and meanings of words. They are often organized alphabetically and may also include information on the pronunciation, usage, and origin of words.
3. **Atlases:** These books contain maps and geographical information. They may also include information on the history, culture, and demographics of different regions.
4. **Almanacs:** These books contain a wide range of information on various subjects, including current events, statistics, and historical data. They are often published annually.
5. **Handbooks and Manuals:** These books provide practical information and instructions on a specific subject or activity. They may include step-by-step guides, diagrams, and illustrations.

Reference books are often found in libraries, schools, and other educational institutions. They can also be purchased for personal use. It is important to choose a reputable and up-to-date reference book to ensure that the information provided is accurate and reliable.

Workshop Practice, H S Bawa, McGraw Hill

- Workshop Practice is a book written by H.S. Bawa and published by McGraw Hill Education.
- The book is in its 2nd edition.
- It is designed for a course on **Workshop Practice** that is offered to all students of engineering.
- This book acts as the basis for further technical studies for manufacturing processes and production technology.
- The book has a total of 250 pages.
- The ISBN 10 of the book is 0070671192 and the ISBN 13 is 9780070671195.

2. Mechanical Workshop Practice, K C John, PHI

- “Mechanical Workshop Practice” is a book written by K.C. John and published by PHI Learning in 2010 .
- The book describes the general principles of different workshop processes such as primary and secondary shaping processes, metal joining methods, surface finishing, and heat treatment .
- The book has 240 pages and has received 4 reviews on Google Books .

3. Workshop Practice Vol 1, and Vol 2, by HazraChoudhary , Media promoters and Publications

- Workshop Practice is a two-volume book set written by HazraChoudhary and published by Media Promoters and Publications.
- The books provide a comprehensive guide to workshop practices and techniques.
- Volume 1 covers topics such as hand tools, measuring instruments, marking and layout, fitting, sheet metal work, and carpentry.
- Volume 2 covers topics such as welding, forging, foundry, machine tools, and electrical and electronic workshop practices.
- The books are written in a clear and concise manner, with numerous illustrations and diagrams to aid understanding.
- They are suitable for students, apprentices, and practicing engineers who wish to improve their knowledge and skills in workshop practices.
- The books are widely used as reference material in technical schools and colleges.

4. CNC Fundamentals and Programming, By P. M. Agrawal, V. J. Patel, Charotar Publication

- CNC stands for Computer Numerical Control.
- CNC machines are automated machines that are controlled by a computer program.
- CNC machines can be used for a variety of manufacturing processes, including milling, drilling, and cutting.
- CNC programming involves creating a set of instructions for the machine to follow.
- These instructions are written in a language called G-code.
- G-code is a standardized language that is used to control CNC machines.
- CNC programming can be done manually, or it can be done using a computer-aided manufacturing (CAM) software.
- CAM software can generate G-code automatically based on a 3D model of the part to be manufactured.
- CNC machines can be programmed to perform complex operations with high precision and repeatability.
- CNC technology has revolutionized the manufacturing industry by allowing for faster and more efficient production.